

AegisCV

AI-powered real-time loss prevention for fashion retail

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PROBLEM: FASHION RETAIL VIEW

Shoplifting causes significant losses in fashion retail

Limited staff visibility



Cameras only record



Loss discovered too late

- staff cannot monitor the entire store
- many stores lack dedicated security staff
- store cameras are mainly used for recording
- suspicious actions are not detected in real time
- theft is often discovered after the customer leaves
- losses are found during inventory checks

- In retail theft “apparel, accessories, and footwear make up the majority of targeted goods at 41.7%.”
Security Systems News / NRF retail crime data (2024)

Fashion is one of the main retail categories targeted by shoplifters.

- “Retailers are losing around €4 million a year in compensation from shoplifters who they are unable to identify.”
DutchNews (2025)

In the Netherlands, shoplifting is already causing measurable financial losses.

Fashion retail is a high-risk theft category with measurable financial impact..

PROBLEM: TECHNICAL VIEW

Existing AI systems are not reliable in fashion retail environments

Research

- Fashion CV still faces **data scarcity, computational constraints, privacy concerns, and algorithmic bias**
Computer Vision for Fashion: A Systematic Review (2025)
Clothing is hard to analyse in stores because garments overlap, change shape, and are often partially hidden by racks or other items.
- Retail vision systems also face **occlusion, theft prevention, and real-time processing** challenges.
A Survey of Challenges and Sensing Technologies in Autonomous Retail Systems (2025)
Fashion stores are visually crowded, so it is harder for AI to track people, items, and suspicious interactions accurately in real time.

AI in fashion is still difficult to deploy reliably in real time.

Industry

- Fashion and luxury companies are using gen AI only “**in select functions for routine tasks.**”
The State of Fashion 2026 (McKinsey × BoF)
This suggests that AI in fashion is still mostly used in limited, lower-risk workflows, rather than in difficult live store operations such as theft prevention
- In loss prevention, accuracy must be “**extremely high ... to prevent false positives.**”
The future of ‘shrink’: the increase of shoplifting (Deloitte, 2024)
In fashion stores, many normal customer actions can look suspicious, which makes false alerts a major problem.

AI in fashion is not yet mature for complex in-store operations.

Current limitations leave fashion retail without a reliable real-time AI theft prevention solution.

SOLUTION: TECHNICAL VIEW

Built specifically for real fashion retail environments

- **Multi-model scene understanding**

We combine transformer-based detection models with optical-flow models to understand customer movement, store dynamics, and object interactions in real time.

- **Context-aware event analysis**

The system does not rely on isolated gestures alone. It evaluates suspicious behaviour in context — including motion patterns, surrounding environment, and event sequence.

- **Store-specific fine-tuning**

Each installation is designed to be fine-tuned during a pilot period for the specific store layout, camera angles, customer flow, and operational patterns.

- **Adaptive false-positive reduction**

After initial suspicious-event detection, the system applies heuristic and neural-network-based filtering to increase or decrease event confidence. Detection thresholds can be calibrated for each store.

- **Extensible detection architecture**

The platform is built to support new algorithms, filters, and behavioural rules over time, enabling adaptation to new theft methods as they emerge.

Our innovation is a context-aware, store-adapted AI system specifically designed for real-time theft prevention in fashion retail, addressing the limitations of generic retail AI.

SOLUTION: FASHION RETAIL VIEW

AegisCV helps fashion stores detect suspicious actions in real time

Full store visibility



Real-time alerts



Prevent loss

- the system analyses store activity in real time
- staff can focus on assisting customers

- suspicious actions trigger an instant alert
- short video clips can be reviewed immediately

- staff can respond immediately to suspicious situations
- losses can be prevented before exit

Operational value

- fewer missed incidents
- quicker staff intervention
- staff more focus on sales

AegisCV turns store cameras into an real-time loss-prevention tool for fashion retail.

PROTOTYPE

AegisCV runs locally and connects to existing store cameras

- **In-store processing**

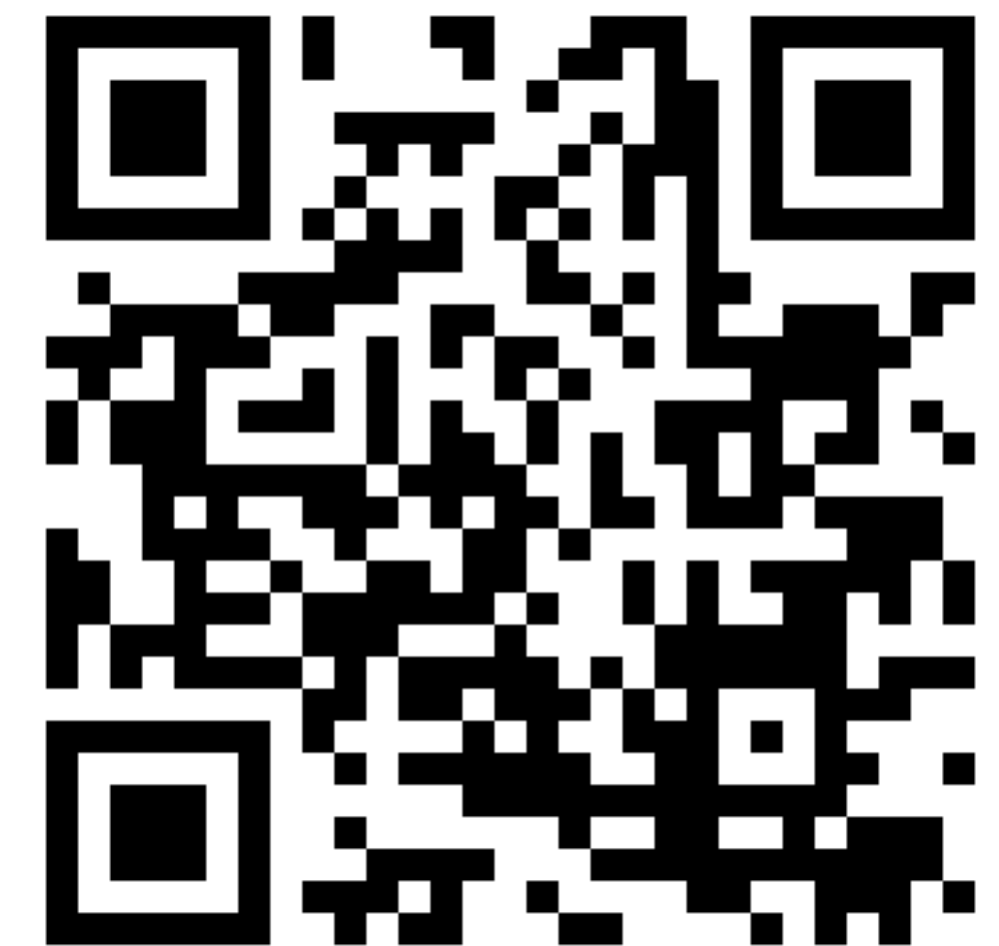
A small in-store server connects to existing cameras and analyses store activity in real time.
The system is configured for each store environment to ensure reliable detection.

All processing happens locally in the store. Video data is stored only for a short time.
No facial recognition or personal identification.

- **Staff notification anywhere in the store**

Store staff receive alerts on a tablet or smartphone connected via Wi-Fi.
Staff can quickly review what happened and decide how to respond.

A working prototype is currently being tested in real fashion store scenarios.
prototype demo: aegiscv.com/demo



Scan to watch demo

MARKET

Most existing retail AI solutions are built for other retail formats

	AegisCV	Veesion	Trigo	Traditional CCTV
Designed for fashion retail	✓	✗	✗	✗
Context-aware analysis	✓	✗	partial	✗
Easy deployment for single stores	✓	partial	partial	✓
Real-time behavior detection	✓	✓	✓	✗
Privacy-friendly (no facial recognition)	✓	✓	✓	✗

AegisCV analyses behaviour in context to support store staff in fashion retail environments.

CUSTOMER VALIDATION

Fashion retail staff confirm the problem and interest in the solution



- **25+ interviews with store managers and shop assistants**
- **most stores report regular concealment incidents**
- **none of the retailers interviewed reported knowing of a comparable solution for fashion stores**

ROADMAP

Pilot deployment in real fashion stores is the next step

Completed



In progress

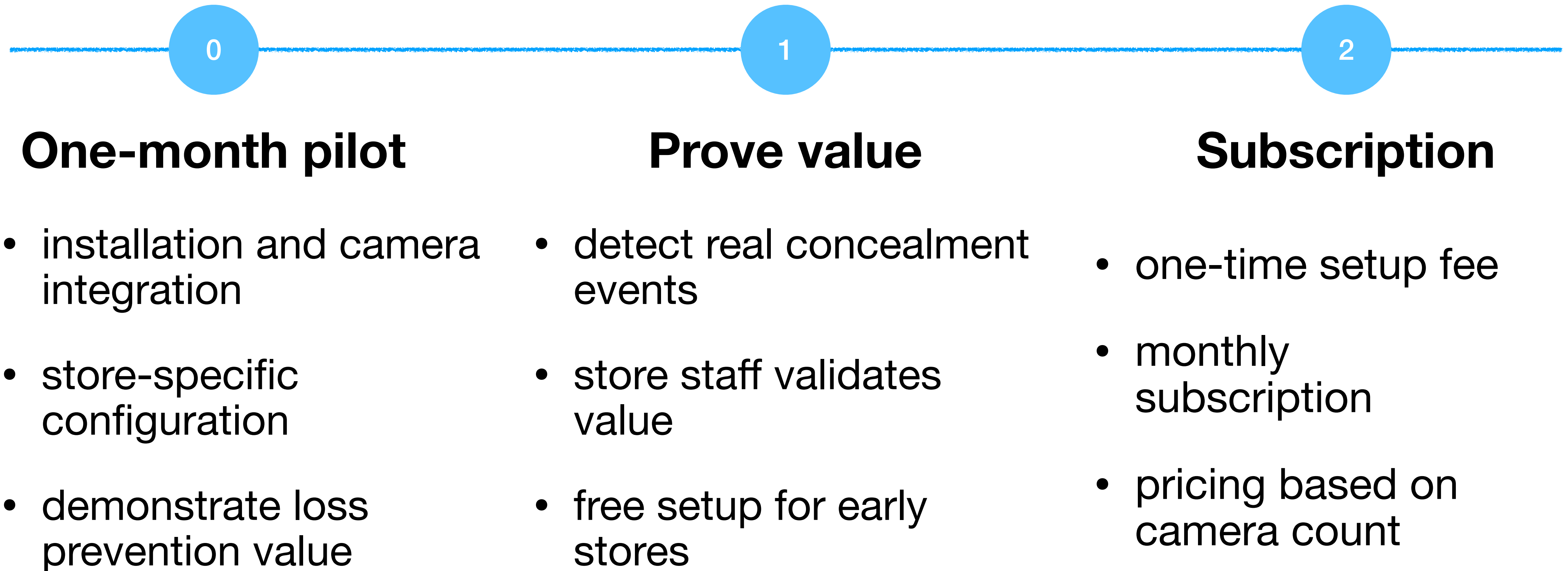


Next step

- | | | |
|--|--|---|
| <ul style="list-style-type: none">• scalable system architecture | <ul style="list-style-type: none">• additional concealment scenarios | <ul style="list-style-type: none">• pilot deployment in 1–3 fashion retail stores |
| <ul style="list-style-type: none">• core concealment detection algorithm | <ul style="list-style-type: none">• model accuracy improvement | <ul style="list-style-type: none">• testing in real store environments |
| <ul style="list-style-type: none">• testing on recorded store video | <ul style="list-style-type: none">• false positive reduction | <ul style="list-style-type: none">• staff feedback and workflow validation |

BUSINESS MODEL

One-time setup fee + monthly subscription



TEAM

20+ years of experience in software development, technology sales and product launches.

Founder & CEO

Head of Architecture (7+ years)

Designed large-scale enterprise software systems

CTO & Co-founder (2 years)

Built integration platforms for retail companies and mobile operators

Founder & CEO (3+ years)

Developed a mobile graphics and computer vision engine

Core team

AI / ML engineer

Neural network models and training

Marketing & design manager

Brand, product design and marketing

WHY THE NETHERLANDS

The Netherlands is a strong starting point for piloting AI solutions in fashion retail.

- Roermond is home to Designer Outlet Roermond, one of Europe's largest outlet centres (200+ stores)
- I live near the Roermond outlet area, giving direct exposure to fashion retail operations and local pilot opportunities
- strong CCTV adoption and developed retail infrastructure support fast deployment in stores
- strict privacy regulations favour privacy-friendly AI approaches
- the compact Dutch market enables fast pilot validation and iteration cycles

CONTACT

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